

Lecture 1

Course Introduction

The course

- About me
- Separate course descriptions
- Content
 - Data communication and computer networks
 - Basic network concepts, network layers and network protocols
 - The structure of the internet and the TCP/IP model
 - Protocols and services of the application layer, the transport layer, the network layer and the link layer
 - Security in computer networks
 - Analyzing network traffic
 - Programming small communication applications



The course

- Goals
 - ..to accept that Computer Networks (especially the Internet) can work.
 - ..to realize how and for what purposes, we can use the Internet.
 - ..to be able to code basic computer systems with a distributed functionality over the internet.
 - ..to be able to design a communication protocol to match the requirements of an application

The course

- The Exam:
 - Oral exam, the Danish 7-point grading scale.
 - Subject presentation + dialog
 - Technical questions requiring technical terms to answer
- **Mandatory exercises** submitted on time and in accordance with the requirements specified during the semester. This applies to:
 - Health Informatics and Technology, Game Development and Learning Technology (Bsc and Msc), Exchange students and anyone else signed up for "Data Communication - T590000102"
 - Failing to hand-in will exclude you from the exam.
- These submissions serve as "løbende evaluering" of the students following "Industrial datakommunikation - T540043102"

The course

Data Communication

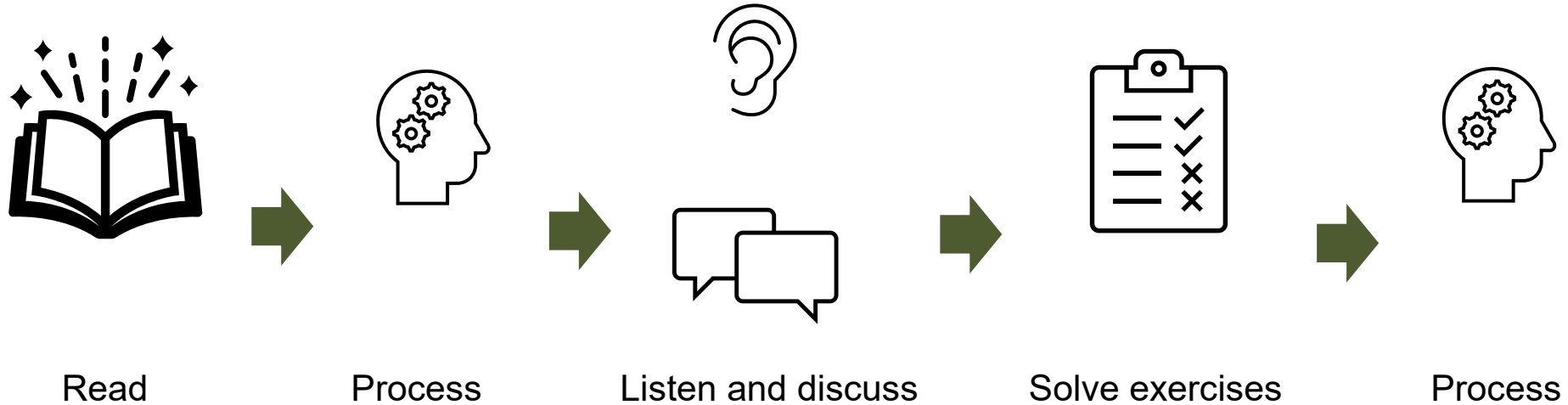


Computer Networks



The Internet

Recommended Approach



Audio and video recordings

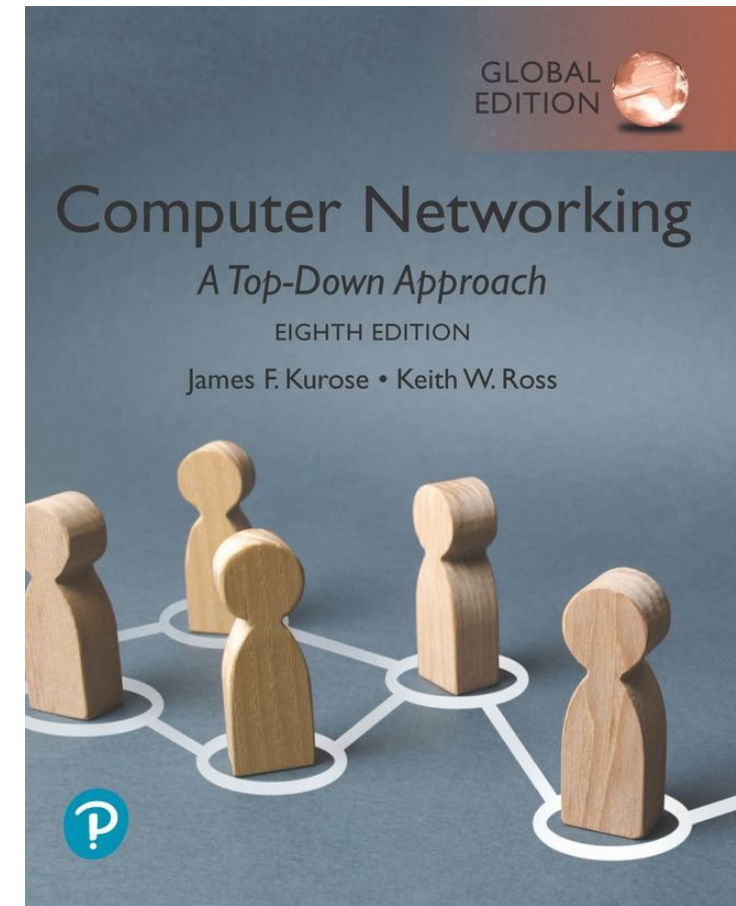
To allow a free discussion in the class, where everybody can ask questions and make comments of all kinds, it is **NOT allowed** to make audio nor video recordings in the classes.

The book

Kurose and Ross
Computer Networking: A Top-Down Approach, 8th Edition
ISBN-13: 978-1-292-40546-9

The Website: https://gaia.cs.umass.edu/kurose_ross/index.php
contains slides and exercises.

9th edition published but not available!



The Plan – See ItsLearning

☐  01 - Introduction LECTURE PLAN Subjects include: Overview of Data Communication & Computer networks No reading prior to the lecture is expected.  7. sep. 08.30–12.00	☐  07 - The Transport layer 2 LECTURE PLAN Subjects: More on TCP Relevant reading: pages 285-314  2. nov. 08.00–12.00
☐  02 - Computer Networks and the Internet LECTURE PLAN Overview of the Internet Recommend reading: pages 31-96 Bring your computer for Lab exercises  14. sep. 08.00–12.00	☐  08 - The Network layer 1 LECTURE PLAN Subjects is the data plane including: IP DHCP NAT  9. nov. 08.00–12.00
☐  03 - The Application layer 1 LECTURE PLAN Subjects: The Application Layer The World Wide Web Mail Protocols Recommended reading: Pages: 111-155  21. sep. 08.00–12.00	☐  09 - The Network Layer 2 LECTURE PLAN Subject is the control plane including: Routing SDN ICMP  16. nov. 08.00–12.00
☐  04 - The Application layer 2 LECTURE PLAN Subjects include: DNS Recommended reading: p111-155  28. sep. 08.00–12.00	☐  10 - The link layer LECTURE PLAN Subjects: MAC ARP Ethernet Relevant reading: page 479-549. I recommend you focus your efforts on reading sections 6.1-6.4. Section 6.7 also serves as a very nice summarization of everything we learned so far.  23. nov. 08.00–12.00
☐  05 - The Transport layer 1 LECTURE PLAN Subjects: UDP TCP Relevant reading: pages 211-285  5. okt. 08.00–12.00	☐  11 - Wireless Communication & IN LECTURE PLAN We also cover an introduction to industrial networking  30. nov. 08.00–12.00
☐  06 - Socket programming in Python LECTURE PLAN	☐  12 - Security LECTURE PLAN Details and reading will follow

Data Communication

Subjects of today:

- Services delivered in data communication
- Transmission structure
- Transmission modes
- Signals and information + examples
- Physical limitations
- First look at protocols
- Network topologies
- peer-to-peer / Multidrop
- Multiplexing
- Circuit vs. packet switching
- Layering and encapsulation + The OSI-model
- Bitrate and propagation delay

Data Communication

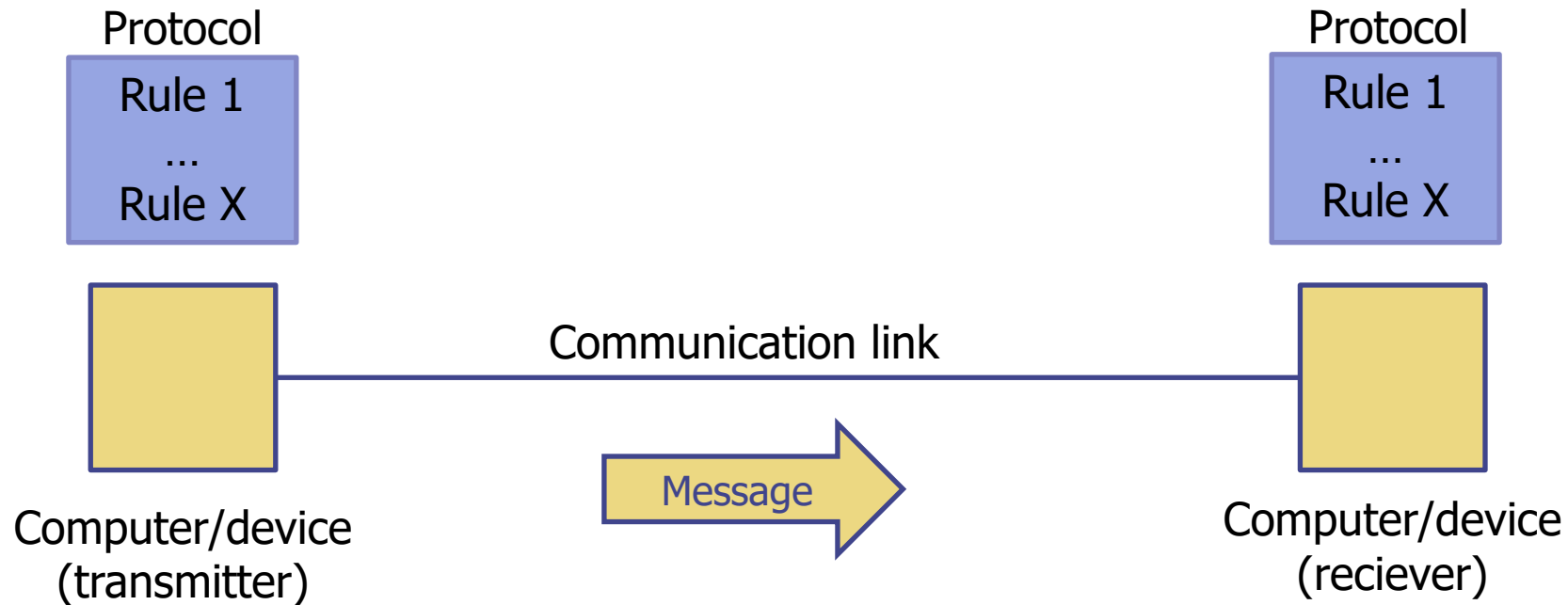
Services?

Data Communication

Services:

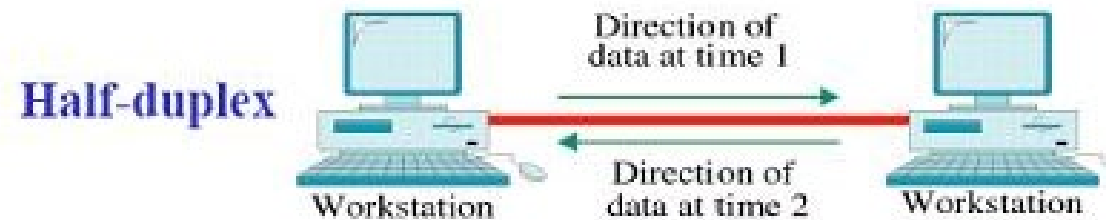
- Delivery
- Accuracy
- Timeliness
- (Jitter)

Data Communication



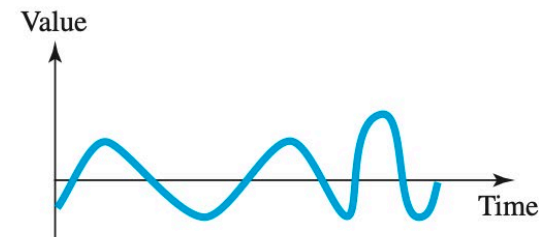
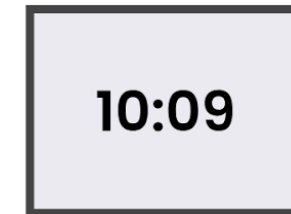
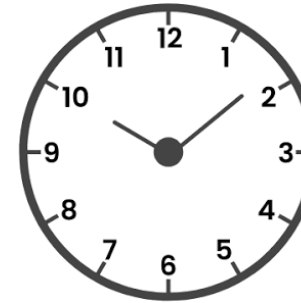
The 5 components of data communication

Transmission Modes

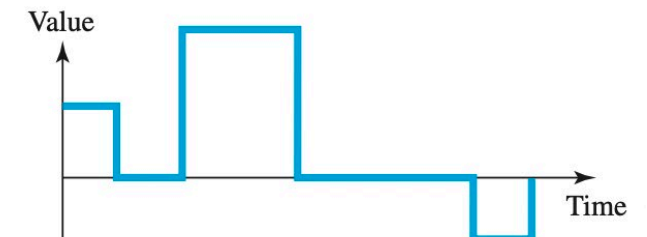


Signals

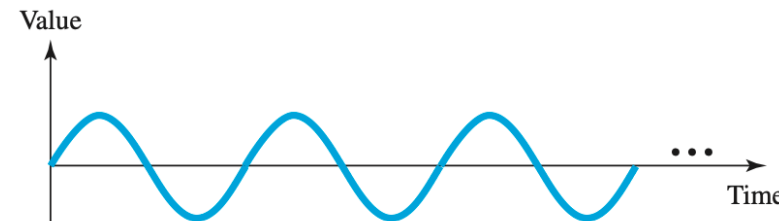
- Analog vs. Digital
 - Continuous vs. discrete
- Periodic and Non-periodic signals



a. Analog signal



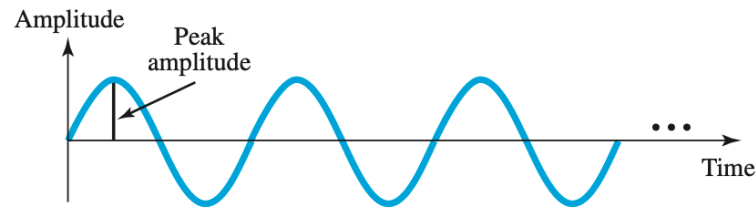
b. Digital signal



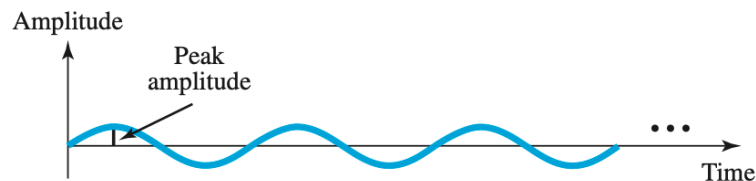
Periodic analog signal

Information in Signals

- Periodic sine waves

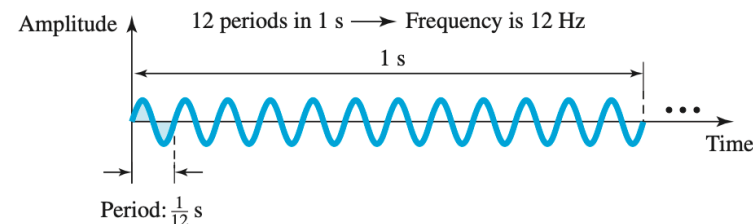


a. A signal with high peak amplitude

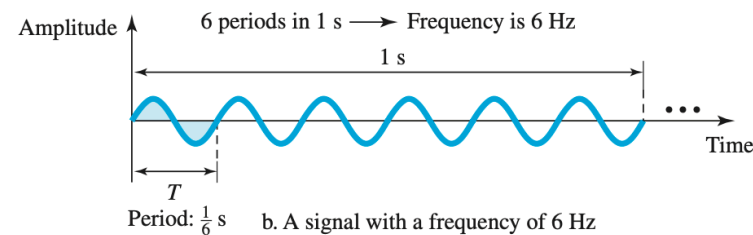


b. A signal with low peak amplitude

Amplitude

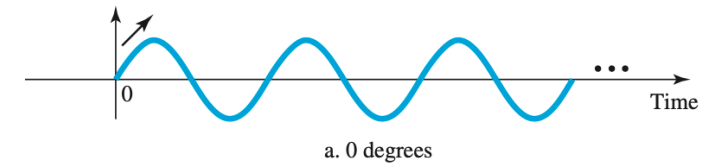


a. A signal with a frequency of 12 Hz

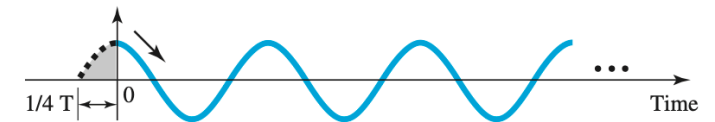


b. A signal with a frequency of 6 Hz

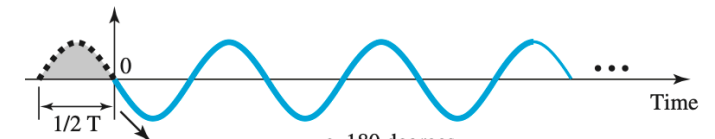
Frequency



a. 0 degrees



b. 90 degrees



c. 180 degrees

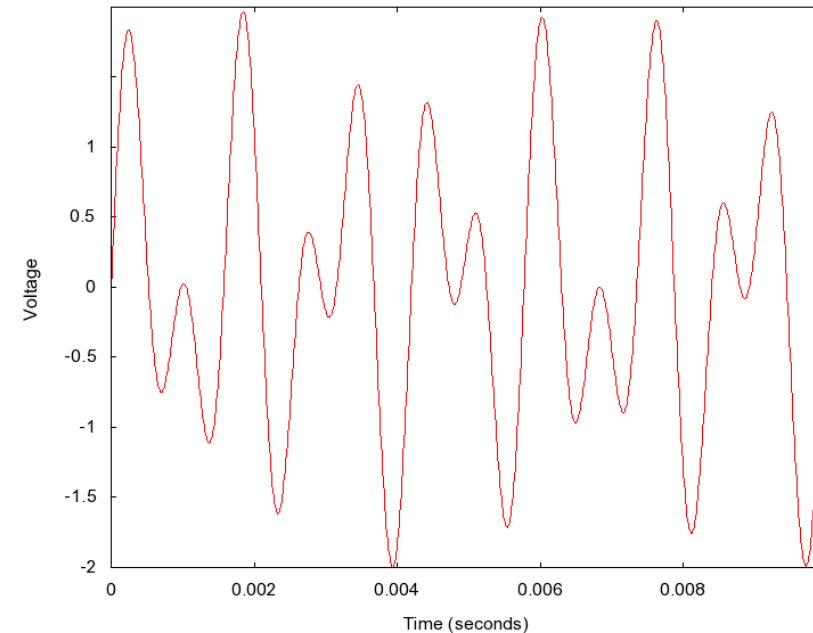
Phase

Dual Tone Multi Frequency

- Press "1" for "Technical support"
- Press "2" for...

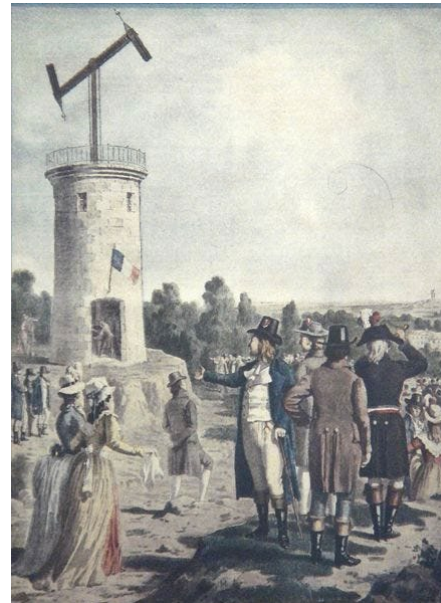
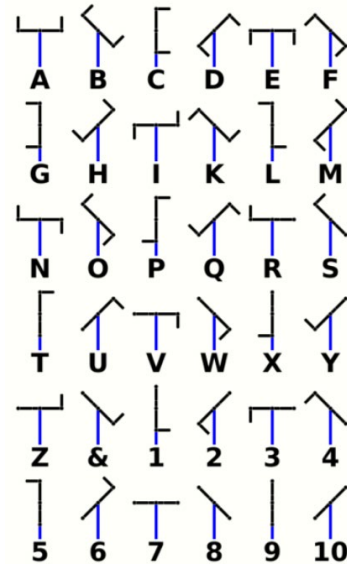
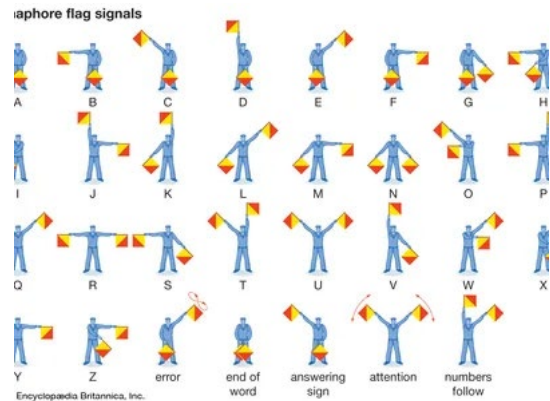
DTMF Keypad

1	ABC 2	DEF 3	11	697 Hz
GHI 4	JKL 5	MNO 6	12	770 Hz
PRS 7	TUV 8	WXY 9	13	852 Hz
* 10	OPER 0	# 11	14	941 Hz
1209	1336	1477	1633	Hz



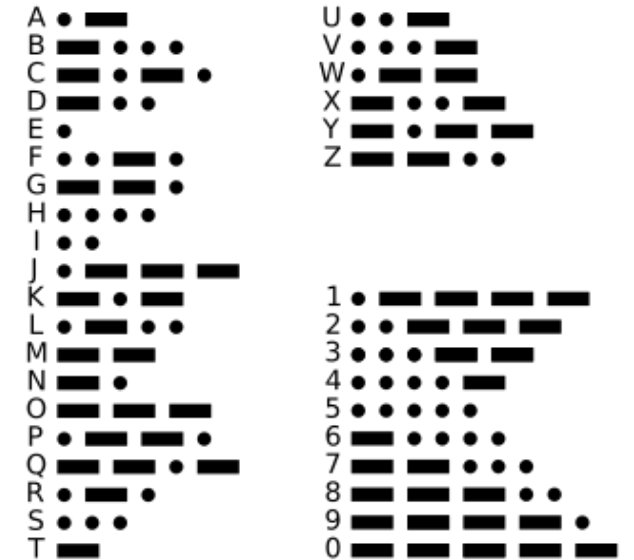
Keypad frequency combinations and an example of a **composite signal** (2 frequencies) and modulation
Source: www.wikipedia.org

Morse encoding



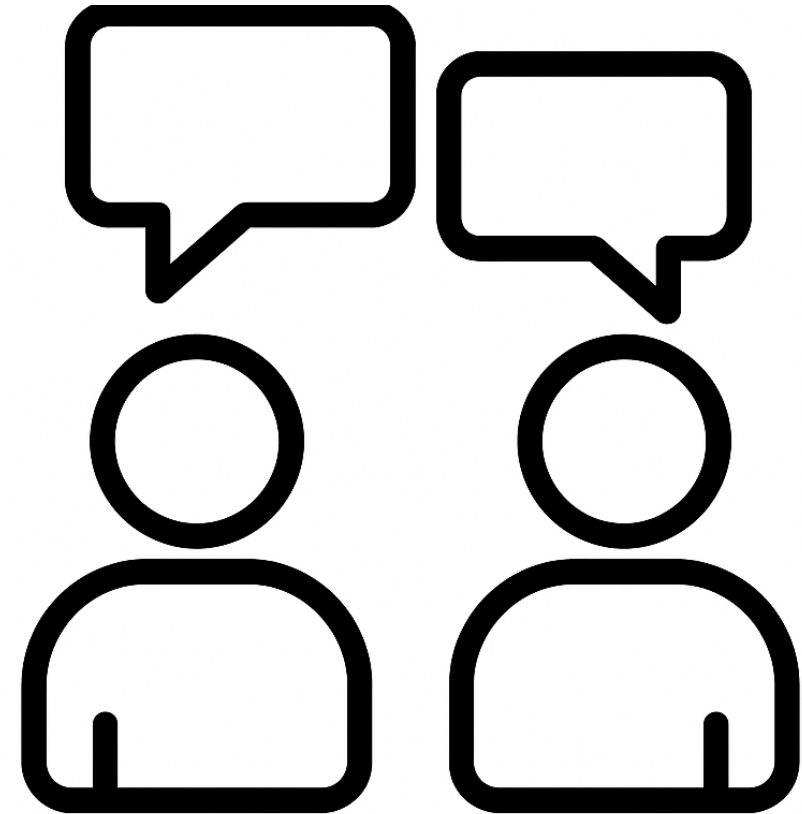
International Morse Code

1. The length of a dot is one unit.
2. A dash is three units.
3. The space between parts of the same letter is one unit.
4. The space between letters is three units.
5. The space between words is seven units.



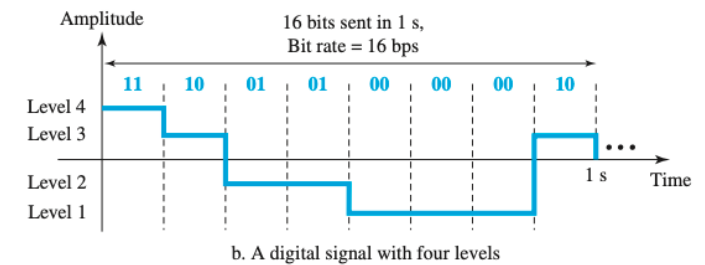
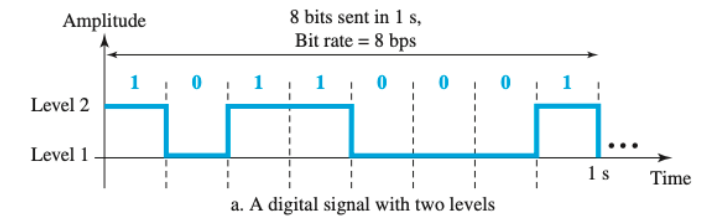
Multi Frequency vs. Morse Encoding

- How does each approach scale if you want to send large amounts of information (e.g. a book)?
Discuss:
 - Transmission time
 - Adding more characters

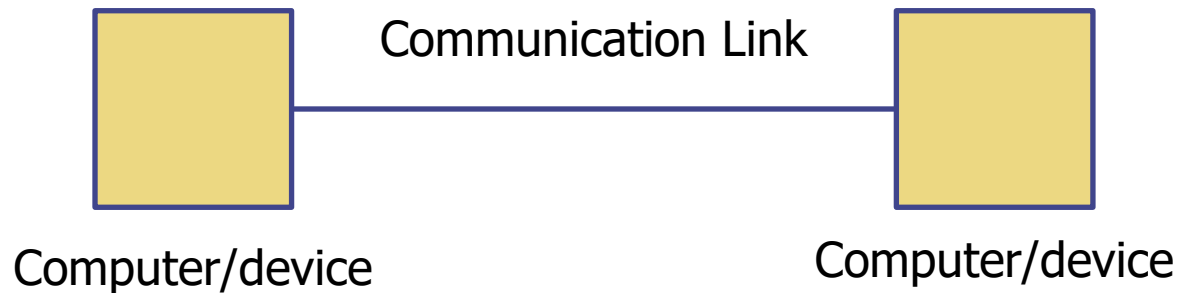


Digital Signals, Bits and bytes

- Non-periodic digital signal and defined logic levels
- Bit is the unit of information (in the computer!)
- 1 bit is the smallest amount of information.
- Information is a measurement of the magnitude of possibilities. E.g.
 - On or off (1 bit)
 - 4 states (2 bits)
 - Numbers (→ more bits)
- Number of possible outcome = 2^n , where n is the number of bits.
- We like to gather bits in chunks of 8, called: bytes.

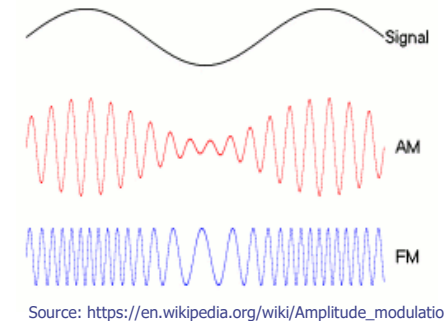


Communicating one bit.

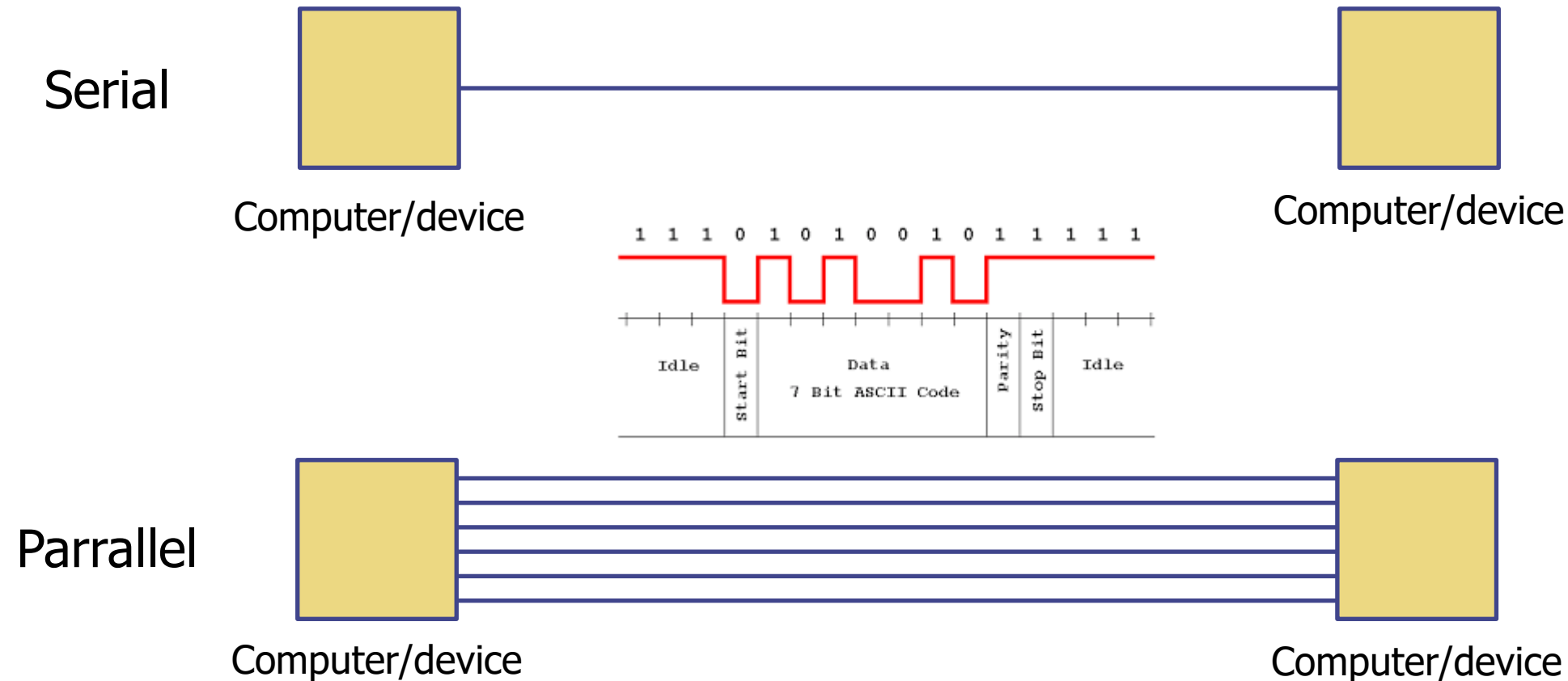


Examples of Logic levels

Link (media)	'0'	'1'
Wire (copper)	0V	5V
	12V	-12V
Radio (air)	AM + ASK	
	FM + FSK	
Light (fiber/air)	Color	
	No light	Light
	Flag down	Flag up

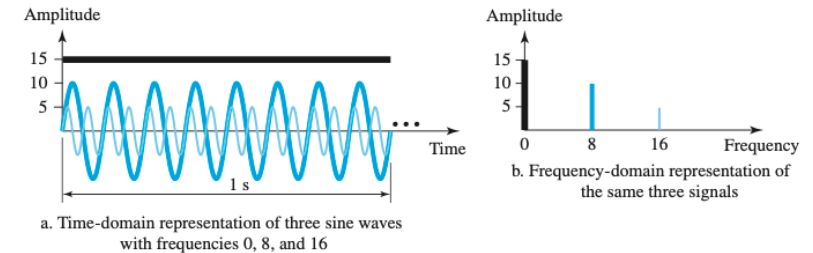


Communicating more information. (more bits).

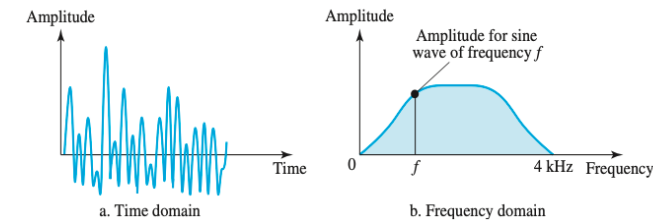


Utilizing Bandwidth

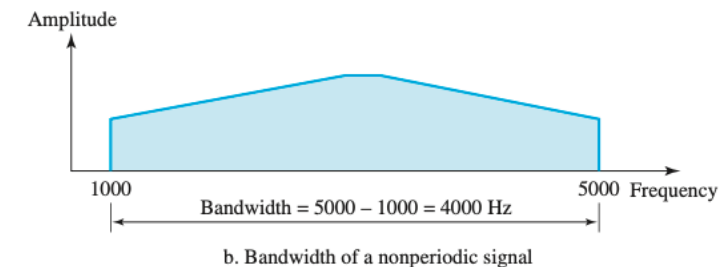
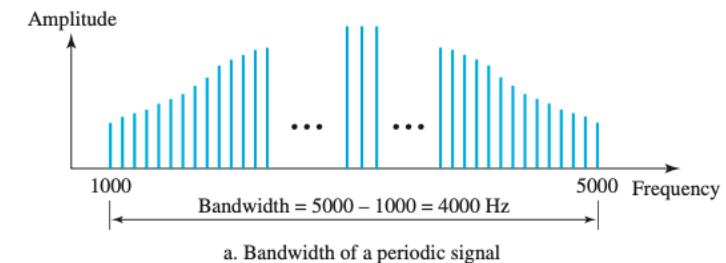
- The band is the range of frequencies that a signal can contain
- The **bandwidth** is the size of this band
- Enables parallel transmissions at different frequencies



3 separate periodic signals

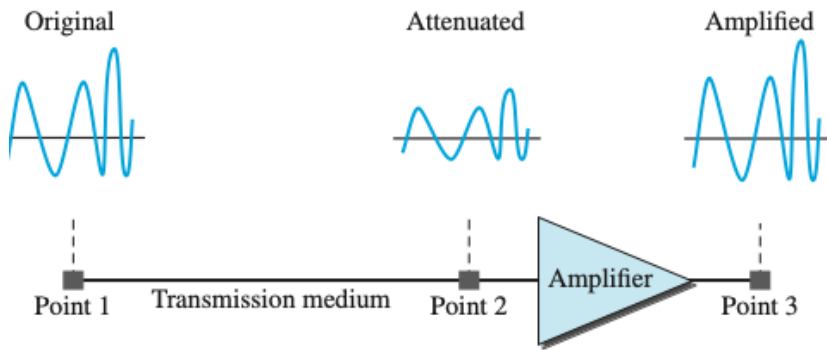


Non-periodic composite signal



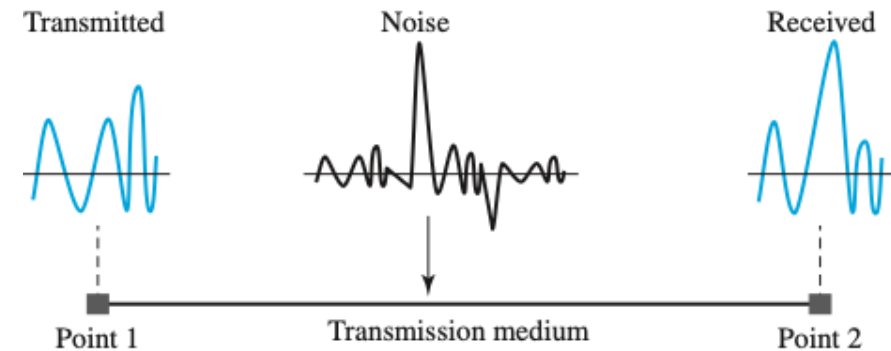
Attenuation and Noise

Attenuation



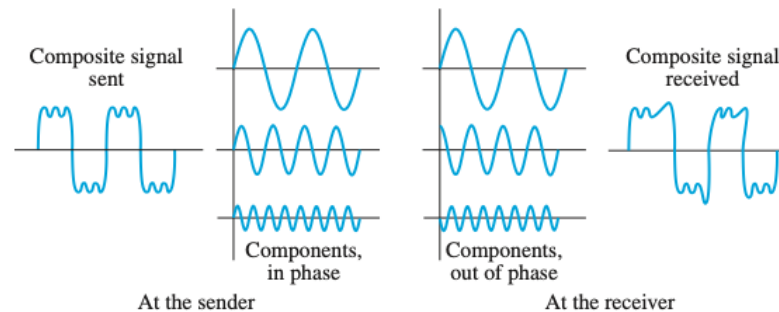
Attenuation: Loss of signal strength

Noise



Noise: Unwanted extra signal

Figure 3.29 *Distortion*



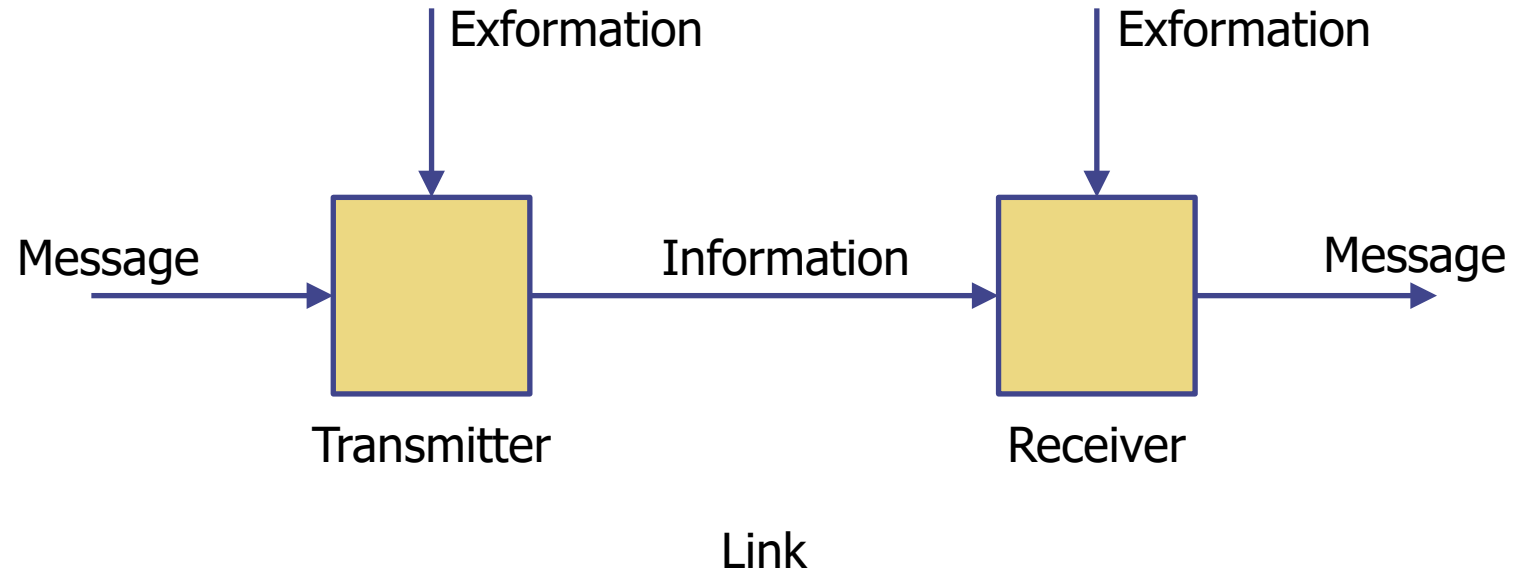
Distortion: A change in signal shape

Course Scope

- Signal processing methods → recover information
- In the course we assume:
 - That a bit stream can be send and received
 - That the end-host CAN handle this flawlessly
 - Errors can happen during transmission
 - There are still problems, but we handle that later!

What is a Protocol?

Information and Exformation



To understand the message, the receiver needs to know the same exformation as the transmitter.
They need to agree on a common protocol of communication.

Binary vs. ASCII

Binary

```
0010 0001
1010 0011
    =
  8611
```

ASCII

```
0011 1000 = '8'
0011 0110 = '6'
0011 0001 = '1'
0011 0001 = '1'
```

The ASCII table

ASCII Codes:

ASCII Hex Symbol	ASCII Hex Symbol	ASCII Hex Symbol	ASCII Hex Symbol
0 0 NUL	16 10 DLE	32 20 (space)	48 30 0
1 1 SOH	17 11 DC1	33 21 !	49 31 1
2 2 STX	18 12 DC2	34 22 "	50 32 2
3 3 ETX	19 13 DC3	35 23 #	51 33 3
4 4 EOT	20 14 DC4	36 24 \$	52 34 4
5 5 ENQ	21 15 NAK	37 25 %	53 35 5
6 6 ACK	22 16 SYN	38 26 &	54 36 6
7 7 BEL	23 17 ETB	39 27 '	55 37 7
8 8 BS	24 18 CAN	40 28 (	56 38 8
9 9 TAB	25 19 EM	41 29)	57 39 9
10 A LF	26 1A SUB	42 2A *	58 3A :
11 B VT	27 1B ESC	43 2B +	59 3B ;
12 C FF	28 1C FS	44 2C ,	60 3C <
13 D CR	29 1D GS	45 2D -	61 3D =
14 E SO	30 1E RS	46 2E .	62 3E >
15 F SI	31 1F US	47 2F /	63 3F ?

ASCII Hex Symbol	ASCII Hex Symbol	ASCII Hex Symbol	ASCII Hex Symbol
64 40 @	80 50 P	96 60 `	112 70 p
65 41 A	81 51 Q	97 61 a	113 71 q
66 42 B	82 52 R	98 62 b	114 72 r
67 43 C	83 53 S	99 63 c	115 73 s
68 44 D	84 54 T	100 64 d	116 74 t
69 45 E	85 55 U	101 65 e	117 75 u
70 46 F	86 56 V	102 66 f	118 76 v
71 47 G	87 57 W	103 67 g	119 77 w
72 48 H	88 58 X	104 68 h	120 78 x
73 49 I	89 59 Y	105 69 i	121 79 y
74 4A J	90 5A Z	106 6A j	122 7A z
75 4B K	91 5B [	107 6B k	123 7B {
76 4C L	92 5C \	108 6C l	124 7C
77 4D M	93 5D]	109 6D m	125 7D }
78 4E N	94 5E ^	110 6E n	126 7E ~
79 4F O	95 5F _	111 6F o	127 7F

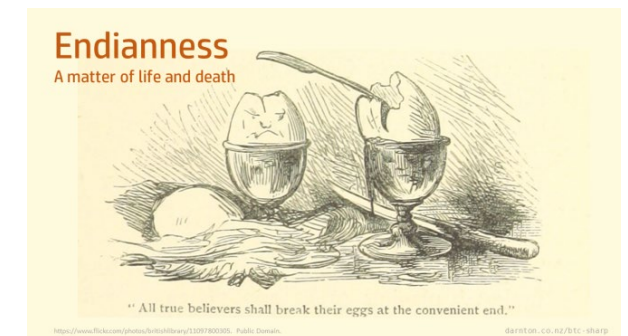
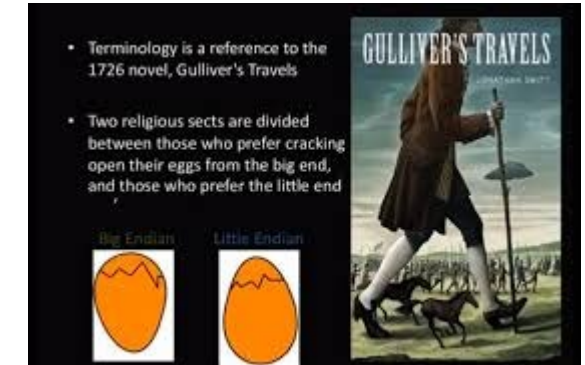
<https://www.ascii-code.com/>

Endianness

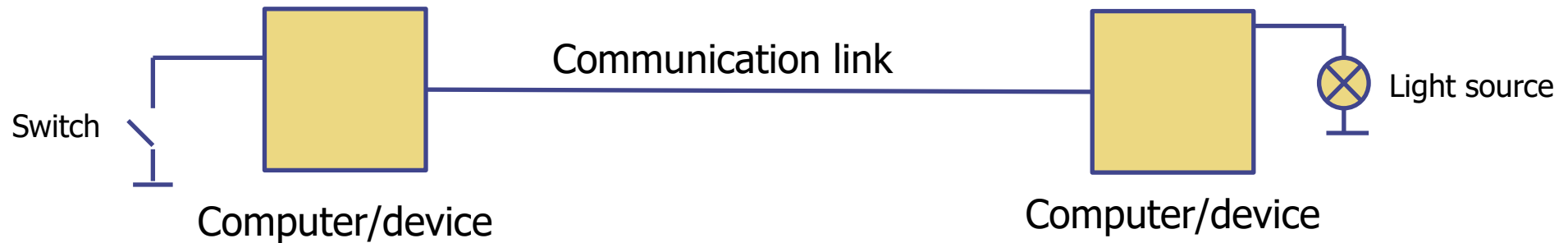
- Big Endian (Motorola a.o.)



- Little Endian (Intel a.o.)



Protocols



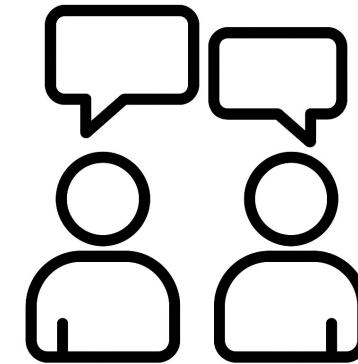
Even at 1 bit information, we need a protocol!

What does it mean?

- Electrically, 0V – 5V
- Mathematically, 0 – 1
- Logically, On – Off
- Functionally, Light – No light

What is a protocol?

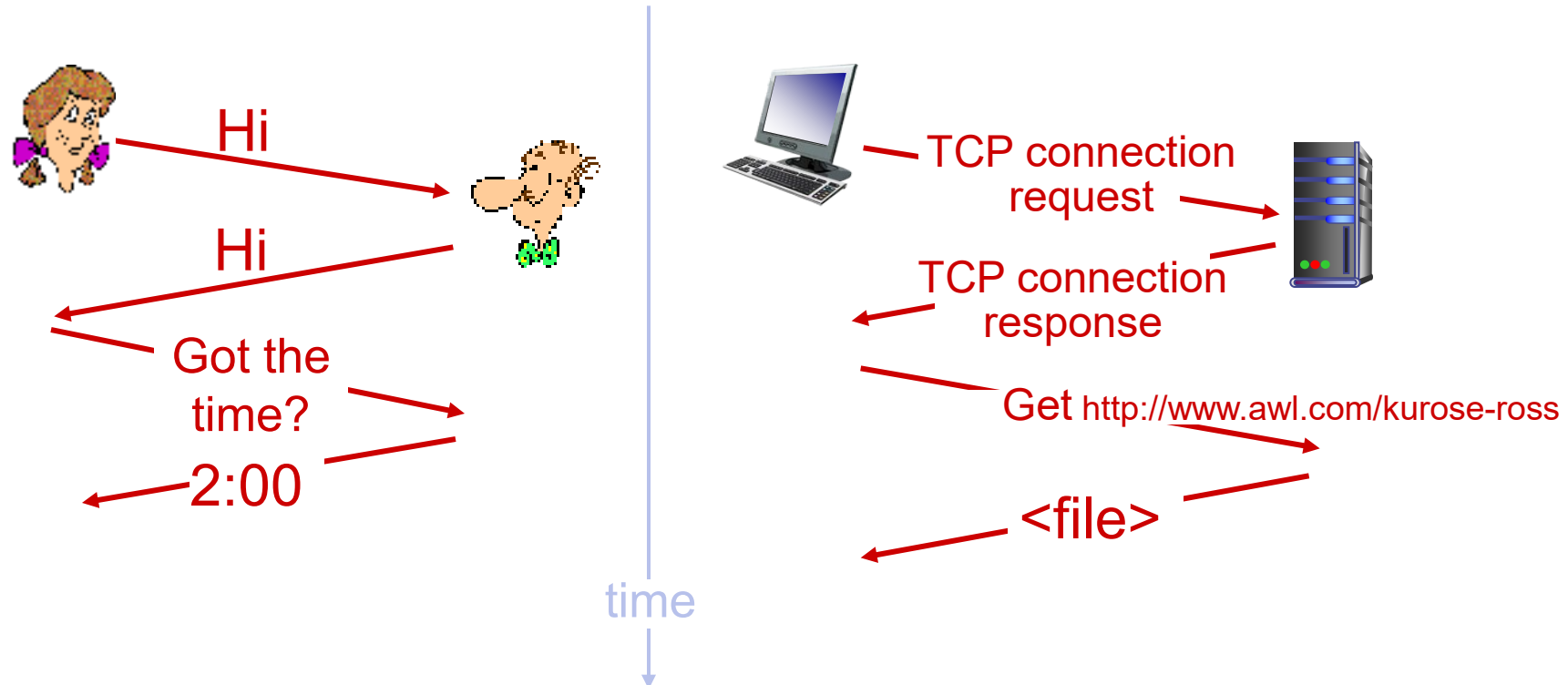
protocols define *format, order* of
messages sent and received
among network entities, and
actions taken on message
transmission, receipt



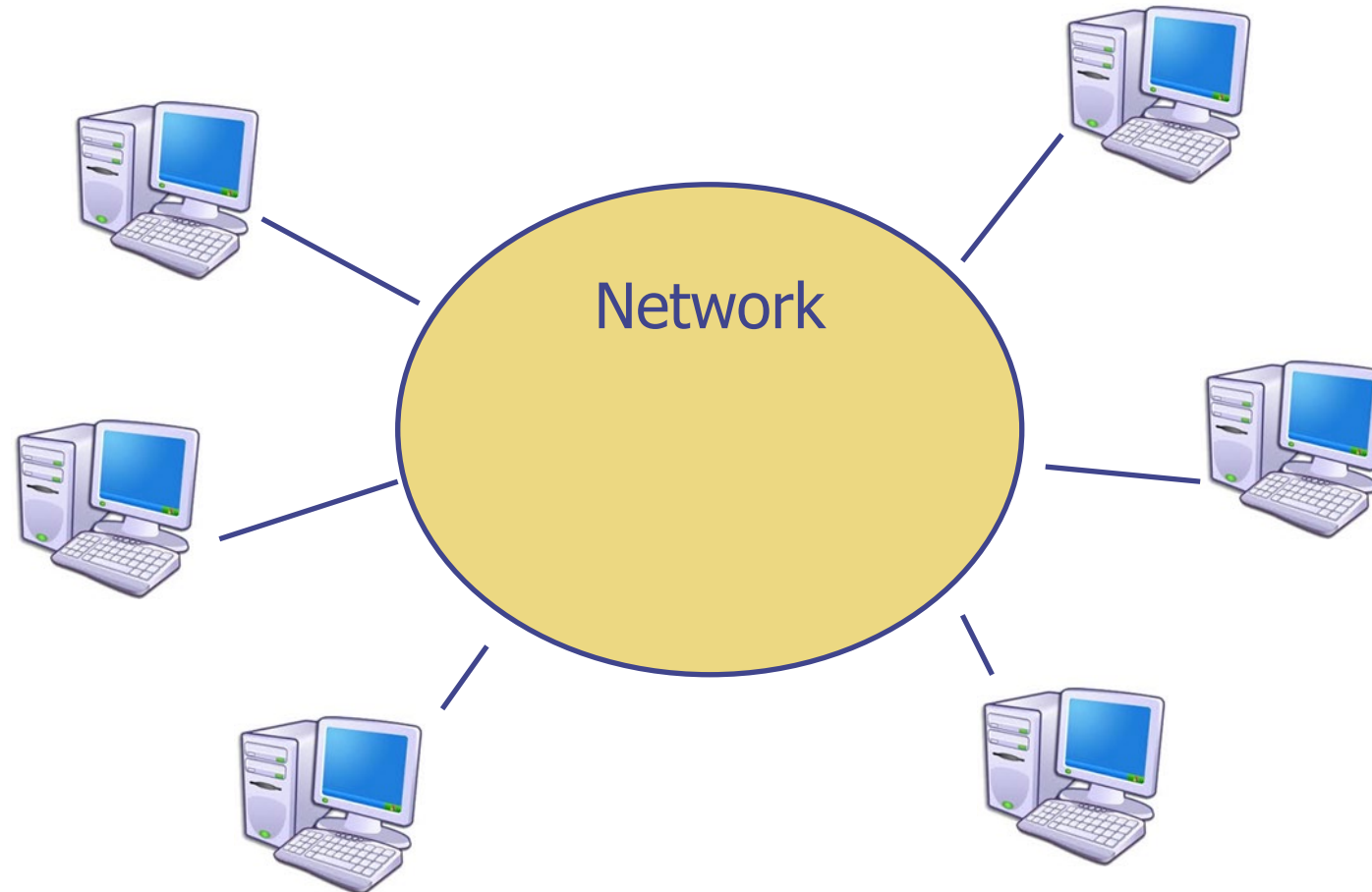
Discuss which
protocols for
humans you
use?

What is a protocol?

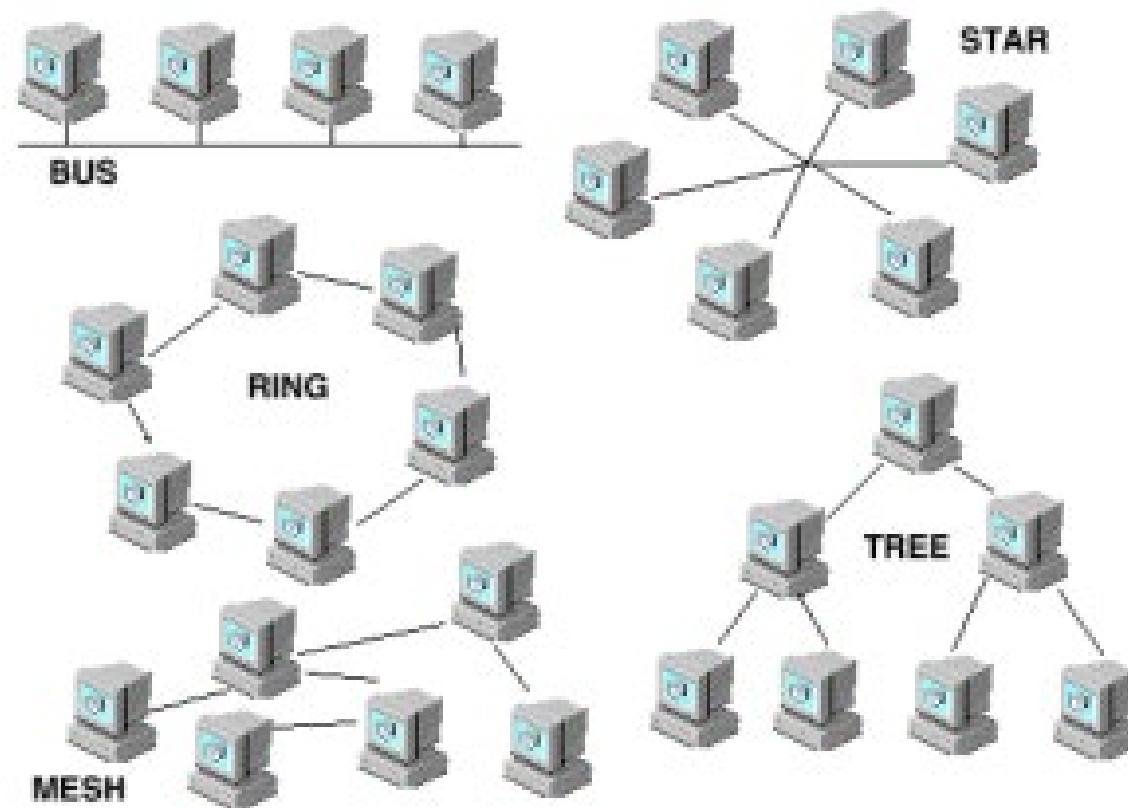
a human protocol and a computer network protocol:



Communicating to more computers/devices.



Network topologies



Peer-to-peer vs. Multidrop

peer-to-peer



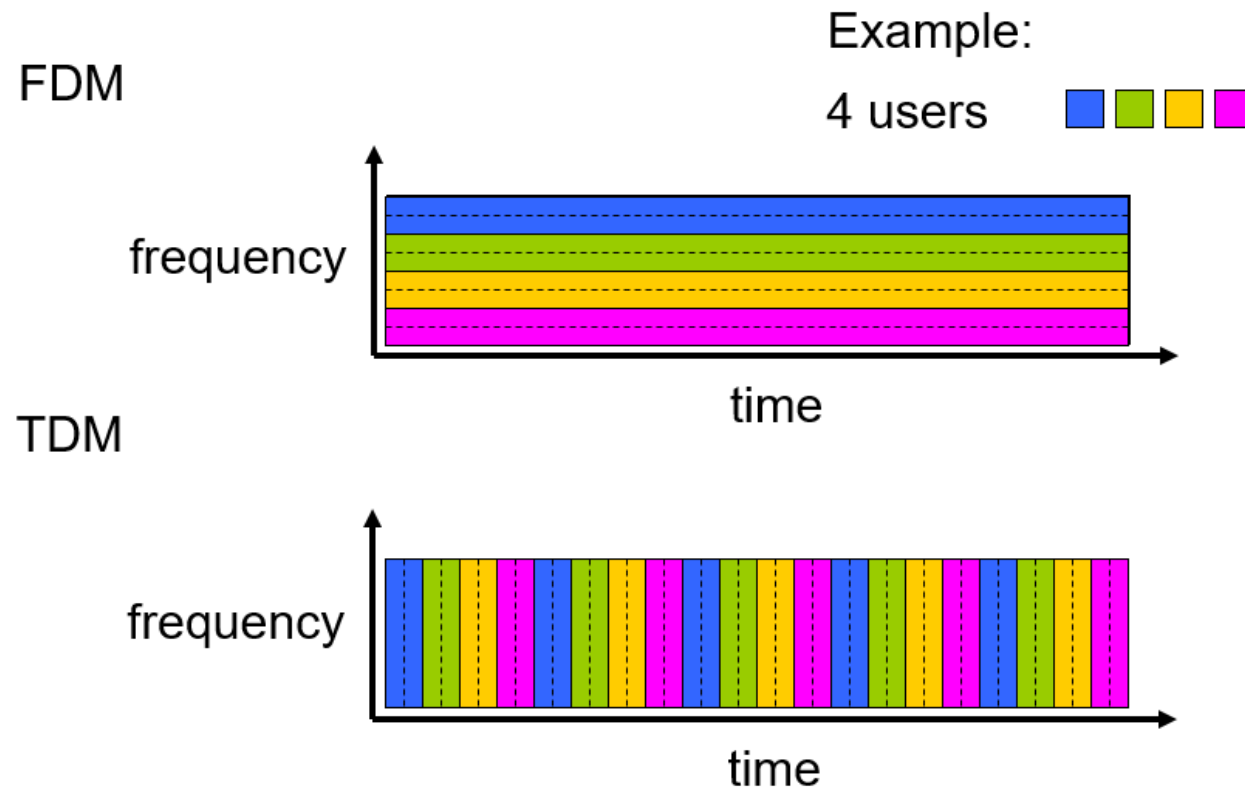
Multidrop



Sharing the communication media - Solutions

- Synchronize.
- Token passing.
- FDM – Frequency Division Multiplexing.
- TDM – Time Division Multiplexing.
- Collision detect and recover.

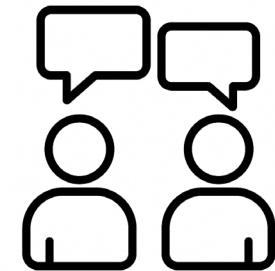
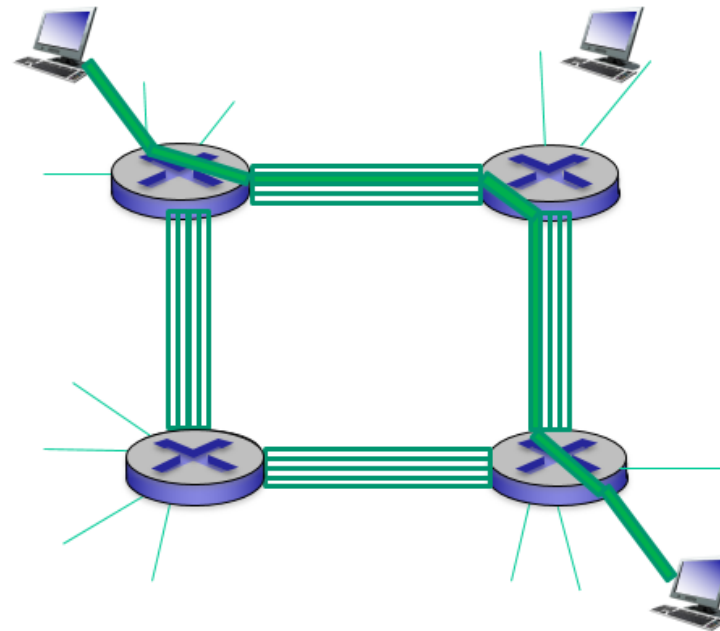
Sharing the communication media - Multiplexing



Circuit Switching

end-end resources allocated to, reserved for “call” between source & dest:

- in diagram, each link has four circuits.
 - call gets 2nd circuit in top link and 1st circuit in right link.
- dedicated resources: no sharing
 - circuit-like (guaranteed) performance
- circuit segment idle if not used by call (*no sharing*)
- commonly used in traditional telephone networks



Why is circuit switching not used for the internet?

Packet Switching

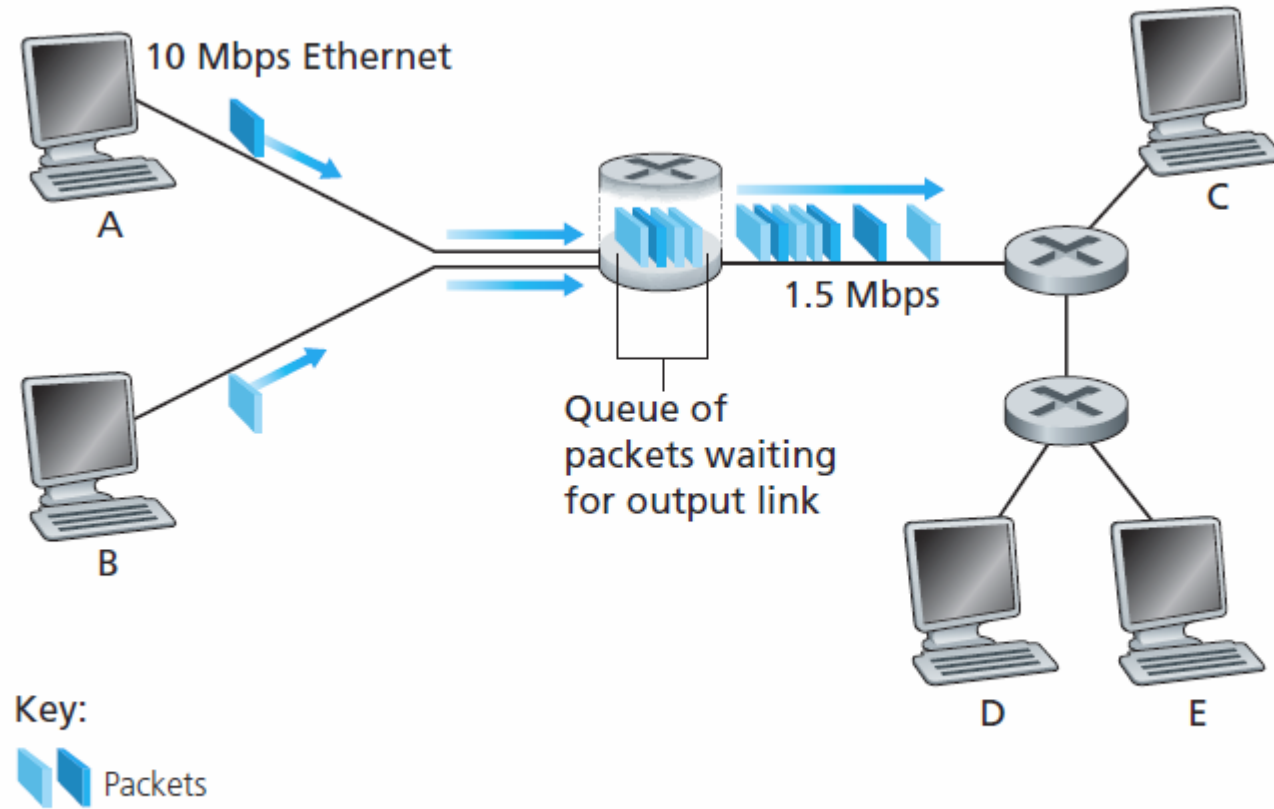
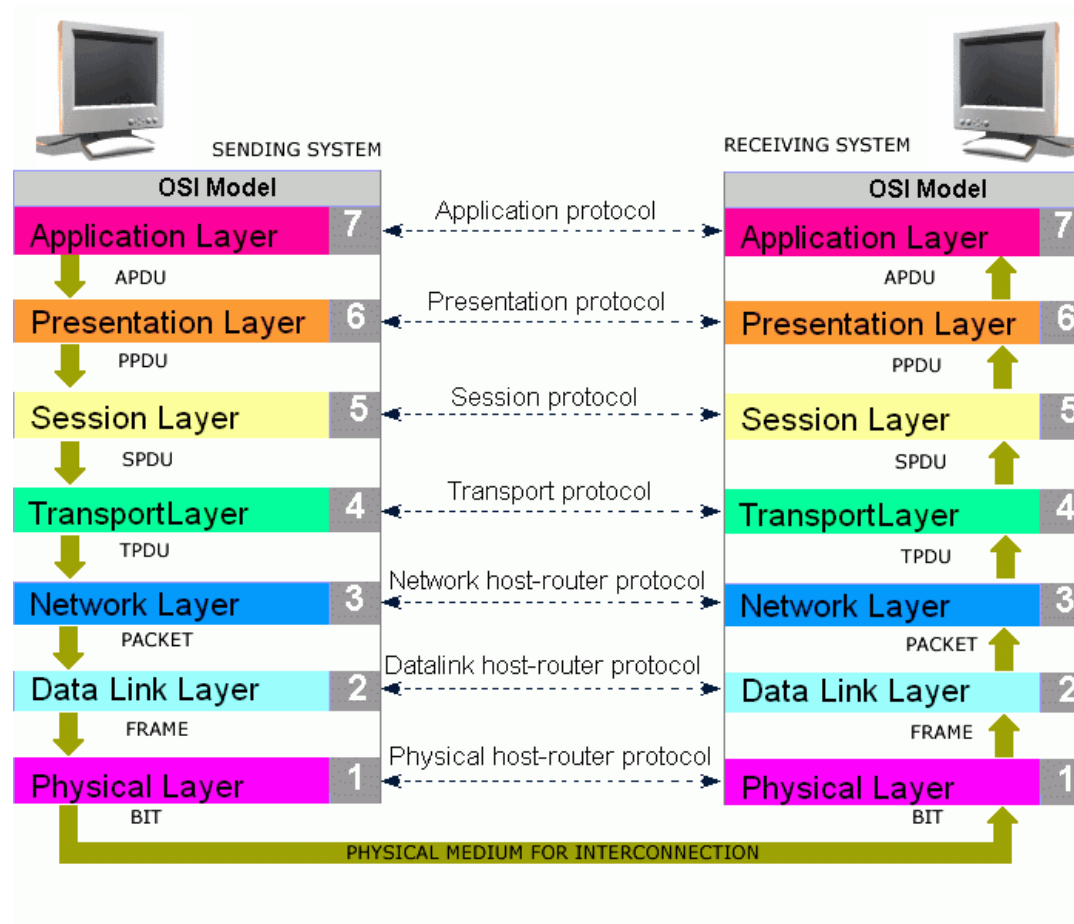


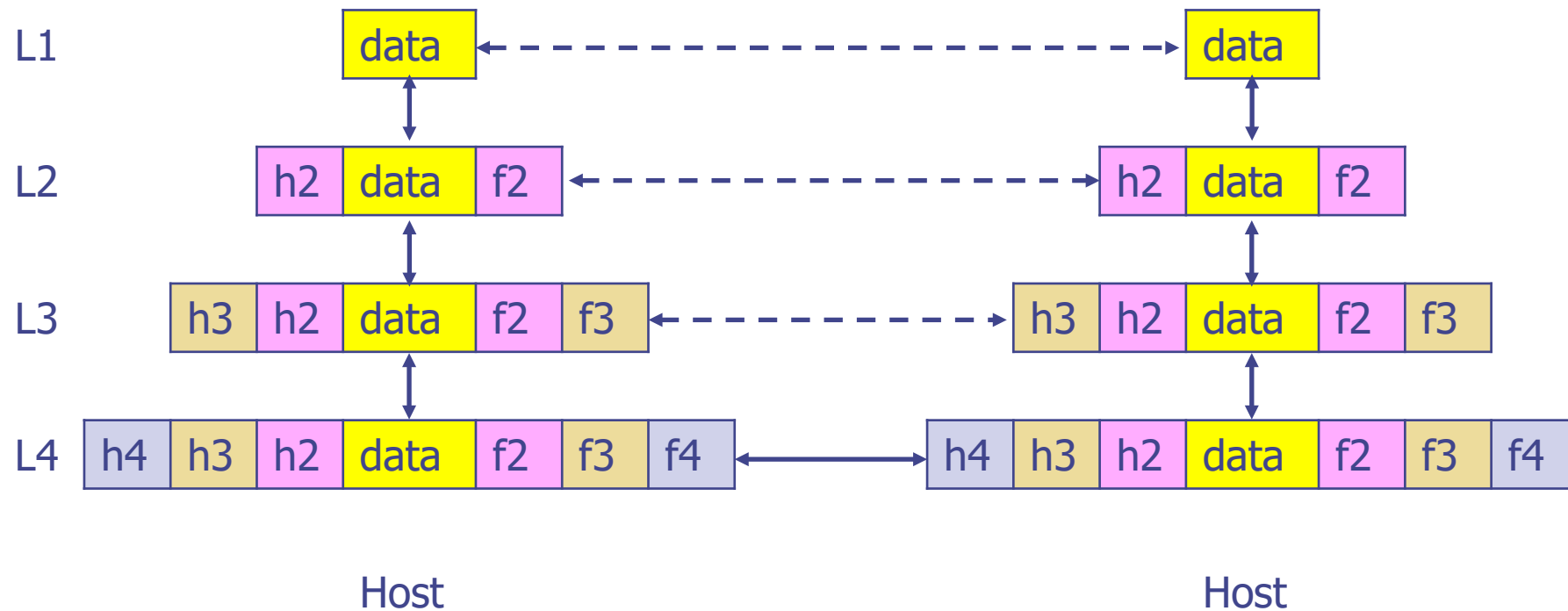
Figure 1.12 ♦ Packet switching

The OSI Reference Model



Open Systems Interconnection
xPDU: x Protocol Data Unit

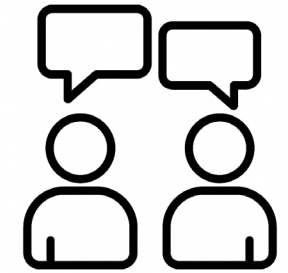
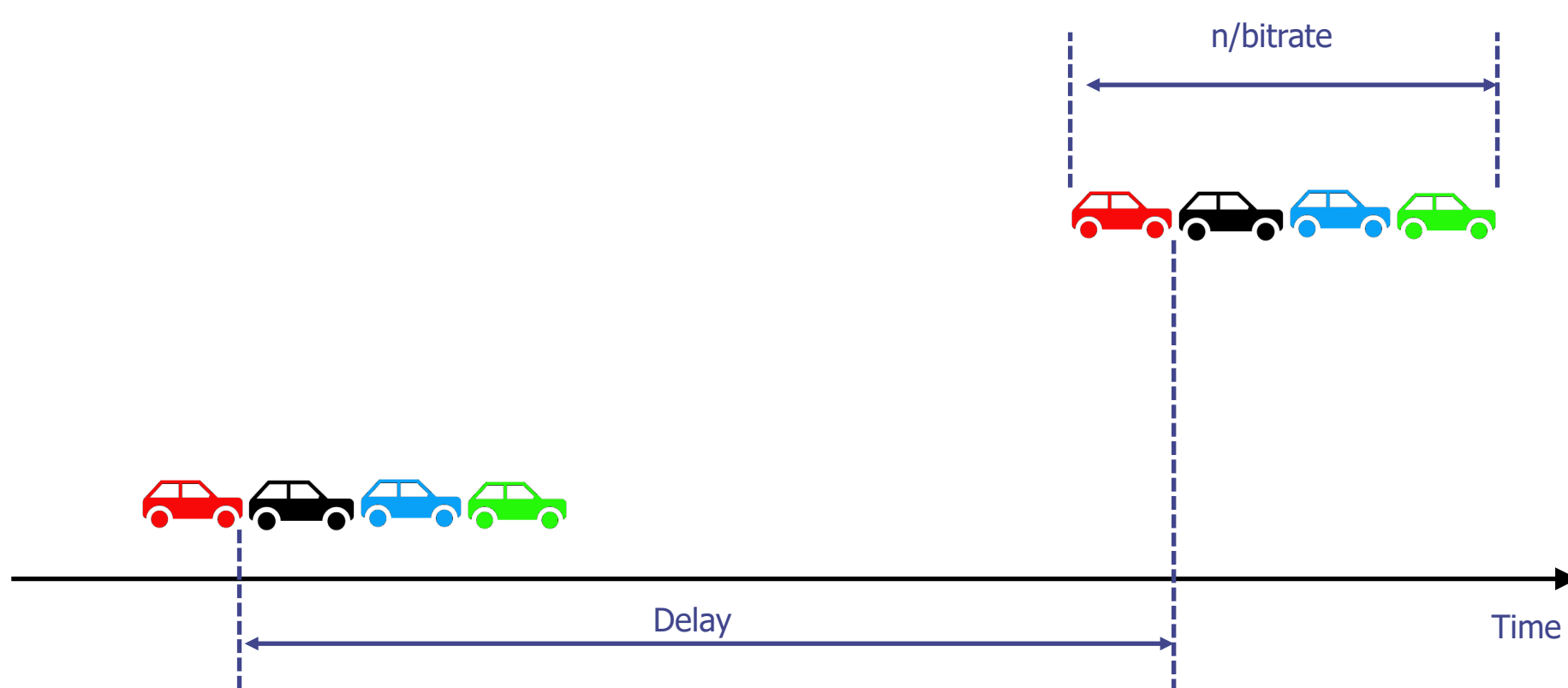
Layering and encapsulation



h = header

Communication speed

Bitrate vs. delay



Which applications on the internet are affected (most) by which factor?

Other Communication Options

- Serial communication
 - RS 232 – Old but still used
 - RS 485 – E.g. industrial communication such as MODBUS RTU
 - SPI communication – e.g. Microcontrollers and peripheral chips
 - I²C communication – Can consist of multi-master and multi-slave setups
- Important principles are covered in the book
- Self-study encouraged!

Next Time

- We talk about Computer Networks and the Internet
- Related reading is Chapter 1 (page 31-96)
- **Bring your computer for Lab exercises**